MYZONE

Shreyansh Soni <sonish@tcd.ie>

some takeaways

3 messages

John Dingliana <John.Dingliana@tcd.ie> 30 July 2026 at 13:13
To: Shreyansh Soni <SONISH@tcd.ie>

Hi Shreyansh,

I think the talk went well. I'm just jotting down below some takeaways that I got from the presentation and discussion while still fresh in mind. All are jsut suggestions, please use your discretion.

- Title "Only the eye of the bird" : while it was engaging for the presentation, I would advise against too "colourful" a title for the dissertation as it may be thought to be contrary to academic convention in our field and ideally the title should be more precisely represent the work to differentiate it or identify it's content if it were housed in a library. You could perhaps keep it as a subtitle.

- The term "window" is a bit ambiguous, though it becomes clearer eventually. Please ensure you define it clearly.

- The name of some of the filters shown seem a bit unintuitive.
Blur and colour displacement would typically be assumed to be colour space operations, but there seems to be specific shape/patterns being formed in the images you showed. Also the grain example has repetitive patterns unlike than what I would expect from a regular whitenoise grain. I'm not sure if this are merely moire effects due to reduced image size in the slides but, if not, I'd suggest reconsidering their descriptions. While this is partly an issue of preference, I feel that from an academic POV you may be seen to be drawing conclusions about somewhat conventional filters using mismatched examples, or that you are not able to implement the filters correctly.

- Minor point: You mentioned in one of the studies that participants encountered sickness due to certain stimuli. Then you went as far as to say it caused them to "puke". This seems a little surprising that an experiment would be allowed to go that far … perhaps this was just a verbal simplification but being sick is different from feeling sick, so please double check if you are going to report this as such.

- you should report the ethics review process; it is useful even to mention the dates of submisison and approval for verifiability but also to show how well perpared you were

- participant demographics should be reported if available (male/female, age, background e.g. computer science students, corrected vision?); if not available then you just have to generalise e.g. they were all above 18 with no known but state the recruitment process so readers can infer something about the sampling

- I think one of carol's questions was related to whether the yielding part of the filter takes into account pedestrian velocity. I guess it doesn't specifically do this. You could just acknowledge this.

- I think another question was related to what happens in the event of too many distractors. You explained this partially in the context of the desktop scenario. But what would happen if there were too many pedestrians and/or road signs in the driving scenario? Perhaps what is suggested/queried is whether there is a budget or prioirtisation of features that would be revealed if there was an overload of distractors, but perhaps there is no perfect solution for this but it should be considered as point for discussion, future work, limitations.

I am available for a final meeting next Tuesday 4th August at 1:45 if you'd like to meet. Note that I am on leave from 11 - 14th so if you have any questions before submission, please let me know before then.

Regards,

John.

Shreyansh Soni <sonish@tcd.ie> 30 July 2026 at 15:37
To: John Dingliana <John.Dingliana@tcd.ie>

Hi Professor,

Thank you for writing these up so quickly.
Tuesday 4 August at 1:45 suits me, and I would like to take it.
Replies in your order below.

1 · Title. Agreed, and that is how it is already set up: the dissertation title is **Adaptive Visual Noise Cancellation for Attention Guidance on Consumer Passthrough Headsets**,
and "Only the Eye of the Bird" was a presentation device only. I mentioned in the introduction slide as well, but didn't mention. If you think the academic title still needs sharpening, it does lean on "visual noise cancellation", which is a borrowed
term rather than an established one, I am happy to work on it on Tuesday.

2 · "Window". You are right that it is doing two jobs. There are genuinely two distinct things and I will name and define
them separately in Chapter 4, at first use:

- the **focus window** - the region the user places and anchors to the room, inside which the application draws nothing at all;
- the **detection aperture** - the smaller openings the detector creates over a recognised light, sign or person, which are transient and not user-controlled.

3 · Filter names. Would it be better if I describe the operation than lean on a conventional name I might be misusing? So in the dissertation I will lead with the mechanism and give the code name in brackets on first mention only, for example "progressive peripheral blur with contrast restoration (Blur)" and "warm–cool chromatic opposition (ChromaticCool)".
On the images themselves: I think most of what you saw is a slide artefact, but not all of it.
The grain mode is deliberately not white noise, it is world-locked static grain at a fixed angular scale, so that the pattern stays put relative to
the room rather than crawling with the head, which is the property the source technique (Cao et al., 2021) depends on. At the size it appeared on the slide that will alias and read as a repeating texture. The blur mode is a genuine two-ring Gaussian, so any structure visible there is compression.

4 · The sickness point. It came from the literature review, and having gone back to the paper my phrasing was wrong.
Norouzi, Bruder & Welch (2018) measured pre/post simulator-sickness questionnaires plus a discomfort rating each minute; their
finding is that vignetting tied to head movement produced a significant *increase* in sickness rather than the intended reduction, and felt unwell due to that.
In my own sessions, one participant felt unwell, took a break and then completed their runs; no session was terminated and nobody was sick. The only data excluded is the run that failed the false-alarm criterion we discussed.

5 · Ethics. I will report the REAMS process properly, with submission and approval dates and the reference number, in the methodology chapter. One question: my understanding is that I should include the blank participant information sheet and consent form as an appendix, and retain the signed originals securely rather than submit them, since they carry personal data. Could you confirm that is what the School expects?

6 · Demographics. Being straight about this: I did not collect demographic data, the post-session questionnaire has no age, gender or vision items, so I cannot report those. Rather than estimate them retrospectively, I will report the inclusion criteria and the recruitment route: all participants were over 18, recruited by direct approach from the postgraduate computer-science cohort, unpaid, with normal or corrected-to-normal vision by self-report, so that a reader can infer the sampling frame and its narrowness. I will list the absence of demographic data as a limitation.

7 · Carol's question on pedestrian velocity : no, it does not. the detection apertures (yielding of filter in selective regions)
respond to a detected object's **angular position and**
apparent size, evaluated per frame, with a minimum-lifetime gate and a smoothing fade. There is no velocity or trajectory term anywhere in either path. A pedestrian stepping off a kerb and one standing still are treated identically if they subtend the same angle. I will acknowledge this explicitly, since it is a real limitation for the scenario the footage depicts: approach speed is exactly the cue a driving-relevant system ought to use.

8 · Overload. There is a hard cap, and it differs by build, which I should be precise about. The live passthrough path allows **8** simultaneous apertures, the video path used for the driving scenario allows **16**. Slots are filled in the order detections arrive and any beyond the cap are simply dropped, so there is a budget, but no prioritisation.
Neither cap was reached in the reported sessions, but that is a property of the footage rather than a design guarantee.
I will cover this in three places: the cap and its first-come behaviour in the system chapter, the absence of prioritisation in limitations, and in future work the obvious remedy : ranking candidates by urgency (eccentricity, apparent size, time-to-contact) and spending the budget on the top few, so that a crowded scene degrades by dropping the least critical rather than the latest. As you say there may be no clean solution; a filter that opens twenty holes has stopped filtering, so at some density the honest answer is to disable it rather than dilute it.

Few things I would like to ask.

* May I send you drafts before you go on leave? I can have a complete draft with you by **Friday 7 August**, which leaves a working day or two before the 11th, I would rather you see the whole argument than isolated chapters, but I will send it in whatever form is most useful.

* Carol said before leaving she'd be excited to go through my presentation as well, so do I have to send it to her?

* And I would welcome your candid view on the presentation beyond the points above. I know I ran over despite cutting it substantially, and I am not sure whether that read as thoroughness or as too much detail or if any section felt rushed, bloated or unclear, that is the most useful thing I could hear before writing the corresponding chapter.

* I also wanted to wanted to discuss with you if you see any future for this particular project or topic adjacent project, from the point of view of academics and / or research papers.
Considering the results are relatively in the positive side.

Thanks again.

Best,
Shrey

John Dingliana <John.Dingliana@tcd.ie> 31 July 2026 at 13:47
To: Shreyansh Soni <SONISH@tcd.ie>

Hi Shreyansh,

> **Tuesday 4 August at 1:45 suits me, and I would like to take it.**

Great. It's already in the calendar. See you then.

> **3 · Filter names**. Would it be better if I describe the operation than lean on a conventional name I might be misusing? So in the dissertation I will lead with the mechanism and give the code name

> in brackets on first mention only, for example "progressive peripheral blur with contrast restoration (Blur)" and "warm–cool chromatic opposition (ChromaticCool)".

Yes that would work. I would prefer if you didn't call it e.g. blur if it is too far from a typical blur kernel in image processing but if you already have generated/rendered figures with these labels it's probably not worth the time renaming it, and just clarifying it in the descriptions should be fine.

> On the images themselves: I think most of what you saw is a slide artefact, but not all of it.

Hopefully this will be obvious in the report. Mainly through MS Team i was seeing repeating patterns in the speckle filters. I assume it is not intended to do this.

> The grain mode is deliberately not white noise, it is world-locked static grain at a fixed angular scale, so that the pattern stays put relative to
> the room rather than crawling with the head, which is the property the source technique (Cao et al., 2021) depends on.

I mentioned "white noise" because you used that term verbally in your presentation at some point. Perhaps you were referring to something else, if so it will be clear in the report. No action required otehrwise.

> At the size it appeared on the slide that will alias and read as a repeating texture. The blur mode is a genuine two-ring Gaussian, so any structure visible there is compression.

Could you send me the videos (or even a screnshot) of the filters? This is mainly to get an idea of the actual filter styles out of interest, but also to check if perhaps Carol and I were both seeing something unintended.
What I was seeing through MS Teams for blur was the blocked out area in quite a strongly stylised appearance, with grey and black lines as if it was an abstract line drawing. This seemed more like an edge enhancement filter like laplacian, rather than a smoothing operation like a gaussian blur.

> **5 · Ethics.** One question: my understanding is that I should include the blank participant information sheet and consent form as an appendix, and retain the signed originals securely rather than submit them, since they carry personal data. Could you confirm that is what the School expects?

Yes a blank version in the appendix, is a good idea and fairly typical.

The submission of the dissertation is electronic and it is not an expectation (and it is not really feasible nor appropriate) for you to submit signed consent forms for assessment purposes. This may also invalidate anonymity.

While not directly likely to be considered in the academic assessment, ethics policy expects consistency with the retention policy and/or any specific measures you stated in the application. For paper-based forms, the school can't centrally help with this so it is usually the supervisor who takes responsibility for retaining and destroys them. Either drop them down to me in Stack B at some point, OR place them in a sealed envelop for my attention (John Dingliana, School of Computer Science and Statistics) in an internal mailbox OR at the school reception in the Orielly building.

> * May I send you drafts before you go on leave? I can have a complete draft with you by **Friday 7 August**, which leaves a working day or two before the 11th, I would rather you see the whole argument than isolated chapters, but I will send it in whatever form is most useful.

Yes I would encourage you to do this, but please note that (a) I will probably be reading 5 other drafts so the earlier the better (b) as your examiner, I can really only provide high level feedback and perhaps a non-exhaustive subset of some specific suggestions (for impartiality and as I can't really be correcting 'my own input') (c) any feedback will be non-exhaustive and it is possible that there can be shortcomings that come to light at the grading stage, even if there was no explicit feedback during review of a draft.

> * Carol said before leaving she'd be excited to go through my presentation as well, so do I have to send it to her?

Sorry I don't recall this being said. Are you sure she wasn't referring to the report rather than the slides? Sometimes I refer to the presentation collectively as what is presented in the report and slides.

It is not generally an expectation and usually examiners are busy enough reading the report and attending the talks amidst other things. But if you're sure she requested it and you are happy forwarding it, it would do no harm … and may in some cases help visually present details (images, videos) that are harder to bring out in the written report.

> * And I would welcome your candid view on the presentation beyond the points above. I know I ran over despite cutting it substantially, and I am not sure whether that read as thoroughness or as too much detail or if any section felt rushed, bloated or unclear, that is the most useful thing I could hear before writing the corresponding chapter.

Some parts were a little rushed and perhaps you were trying to still do too much in a 30 minute talk but you were clearer than the other students that I sat through on the day. The presentation clearly showed a lot of thought and effort went into it and that you were committed and engaged with the project.

The report should naturally be more detailed i.e. you don't need to streamline this based on the presentation/demo, in fact you should be clarifying or adding in any information that was removed from the presentation, due to time constraints of the talk.

> * I also wanted to wanted to discuss with you if you see any future for this particular project or topic adjacent project, from the point of view of academics and / or research papers.
> Considering the results are relatively in the positive side.

I think it is definitely an interesting area of work worthy of further development refinement given time and resources, and I also think that it has potential from a practical standpoint i.e. while I fear actual use in driving may be a bit too far down the line due to safety concerns, it seems it could be useful in some of the other use cases discussed or in simulation-based training applications.

REgards,

John